

Yet Another .NET deserialization

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Insecure Deserialization

Yet Another .NET deserialization

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This is my second post on white-box analysis but for another technology stack and vulnerability category: **.NET deserialization leading to Remote Code Execution**. The outcome for this self-paced training was *CVE-2019-18211 *with an outstanding product vendor (friendly/fast communication and quick bugfixes available since 30th October 2019)**.

After installation of the latest C1 CMS package (back then v6.6) via Microsoft Webmatrix installer, a running instance was dynamically investigated with live debugging (using: **dnSpy**). The standard administrator account of C1 CMS was used to add several *lower privilege accounts* e.g. holding Editor and Developer roles.

A bottom-up approach was used to directly search for well-known .NET deserialization issues, namely looking for user-controlled input resulting in insecure deserialization. C1 CMS seemed to *prefer JSON objects* being sent through SOAP requests. No unsafe deserialization for these kind of objects were found (e.g. TypeNameHandling settings). Technically, deserialization candidates were searched simply by looking for **Deserialize methods** of loaded .NET assemblies. In the following, we focused on the **TreeServiceFacade** and `**EntityTokenSerializer**` class.

Interestingly, a legacy deserialization path was provided if the incoming object is **not based on JSON**. The **serializedEntityToken** parameter in **EntityTokenSerializer** was parsed with respect to a regular expression:

```
_keyValuePairRegex = new Regex(@"\s*(?<Key>[^\s]*)\s*=\s*(?<IsNull>null|(?<Value>[^\s\r\n]*(\\.([^\s\r\n]*)*))\s*,*\s*, RegexOptions.Compiled)
```

Starting from this regular expression, valid objects could be built manually. Now, the **concrete vulnerability** happened to be in the non-JSON part of the class *EntityTokenSerializer*. With the parameter *entityTokenType* one can **fully control which .NET assembly will be used as a type for deserialization**. Afterwards, the code searched for a *Deserialize* method for this assembly name and **invoked this method via Reflection**.

The only restriction for this was that the serialized object came in as a **String**. Looking for a proper *Formatter* lead us to the well-known **BinaryFormatter** which was used under the hood from

Microsoft.Practices.EnterpriseLibrary.Logging.Formatters.BinaryLogFormatter.

So using the *Deserialize* function of this assembly allowed a String parameter which was *base64 decoded* and afterwards fed into the BinaryFormatter deserialization call.

All what was left was creating a malicious payload with the tool **ysoserial.NET** (<https://github.com/pwntester/ysoserial.net>). As a proof-of-concept the following command was used:

```
ysoserial.exe -g TypeConfuseDelegate -f BinaryFormatter -c "cmd.exe /C echo pwned > C:\Users\Public\Downloads\PWned.txt" -o base64
```

Delivering the following request as a user with *Editor role* (**low privilege user**) containing `entityTokenType='Microsoft.Practices.EnterpriseLibrary.Logging.Formatters.BinaryLogFormatter'` and `entityToken=BINARYFORMATTERPAYLOAD*` *lead to a Remote Code Execution on server-side*.

Basically, the role didn't matter. **Any authenticated user** was able to achieve Remote Code Execution within the CMS console this way.